

VERESHCHAGINO KRASNOYARSK, Russian Federation

WMO: 235780

Lat: 64.236N

Lon: 87.573E

Elev: 99

StdP: 14.64

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-50.0	-45.1	-54.6	0.2	-50.1	-50.1	0.3	-45.3	16.8	2.1	14.4	4.6	1.8	180	0.676

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	83.0	64.8	79.5	63.4	75.6	62.9	68.0	76.8	65.9	74.6	64.0	72.4	5.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.6	91.6	72.4	62.6	85.2	69.7	60.7	79.7	68.6	32.5	77.3	30.8	74.5	29.4	72.2	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.8	11.6	DB	-53.6	87.9	4.7	3.7	-57.0	90.6	-59.7	92.7	-62.3	94.8	-65.7	97.5	(4)
(5)			WB	-54.5	70.6	5.1	3.1	-58.1	72.8	-61.1	74.6	-63.9	76.3	-67.6	78.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	-14.4	-5.1	12.3	25.3	36.8	58.8	63.8	55.4	43.5	26.0	-1.1	-4.5	(6)	
	DBStd	29.08	18.25	15.24	14.31	11.38	8.98	9.53	6.79	6.93	7.91	10.48	16.54	17.92	(7)	
	HDD50	10123	1995	1543	1169	740	422	31	1	28	227	744	1532	1691	(8)	
	HDD65	14751	2460	1963	1634	1190	876	229	104	303	646	1208	1982	2156	(9)	
	CDD50	965	0	0	0	0	12	295	429	196	32	1	0	0	(10)	
	CDD65	116	0	0	0	0	1	42	67	6	0	0	0	0	(11)	
	CDH74	1081	0	0	0	0	5	438	597	40	1	0	0	0	(12)	
	CDH80	238	0	0	0	0	1	101	135	1	0	0	0	0	(13)	
	WSAvg	5.1	4.9	5.2	5.6	5.9	5.5	5.0	4.7	4.5	4.7	5.3	4.4	5.4	(14)	
	PrecAvg	31.1	2.1	1.7	1.9	2.0	2.3	2.4	2.7	3.3	2.9	4.1	3.1	2.7	(15)	
(16) Precipitation	PrecMax	39.3	4.4	3.1	3.3	3.0	3.9	5.5	5.3	6.6	5.2	7.2	5.4	4.4	(16)	
	PrecMin	20.5	0.4	0.7	0.3	0.8	1.0	0.8	1.1	1.0	1.0	1.2	1.3	0.8	(17)	
	PrecStd	5.3	0.9	0.6	0.8	0.6	0.7	0.9	1.0	1.2	1.1	1.3	1.0	1.1	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.3	30.3	42.1	52.6	69.6	86.5	87.6	78.2	71.7	53.2	33.0	30.1	(19)	
		MCWB	22.1	29.2	34.0	41.5	54.9	66.0	66.6	64.9	58.4	46.5	32.2	29.2	(20)	
	2%	DB	20.2	24.3	37.2	47.7	61.8	83.1	84.0	74.6	64.4	44.8	29.5	27.1	(21)	
		MCWB	19.1	22.8	31.8	37.8	49.7	63.8	65.9	63.0	56.1	42.3	28.4	26.2	(22)	
	5%	DB	15.6	18.7	34.3	43.7	54.5	79.6	80.6	71.0	58.8	41.4	26.6	24.5	(23)	
		MCWB	14.6	17.5	31.2	36.1	44.6	61.8	64.4	61.7	51.9	39.1	25.6	23.5	(24)	
	10%	DB	11.0	14.6	31.6	40.0	49.5	75.3	77.1	67.6	54.6	37.7	23.0	19.3	(25)	
		MCWB	10.0	13.5	28.9	34.2	42.3	61.8	63.8	60.3	50.3	35.2	22.1	18.3	(26)	
	0.4%	WB	22.5	29.3	35.1	42.4	56.8	68.4	71.2	68.4	60.9	46.5	32.4	29.4	(27)	
		MCDB	23.1	30.1	38.3	47.6	67.1	81.1	81.5	75.2	67.4	51.4	33.0	29.7	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	19.1	23.1	33.2	39.3	50.8	66.0	69.3	64.9	56.6	42.9	28.5	26.1	(29)	
		MCDB	20.2	24.1	36.2	45.1	60.4	77.3	79.7	70.5	63.4	44.5	29.3	26.8	(30)	
	5%	WB	14.6	17.7	31.5	37.0	46.6	64.4	67.3	62.8	53.5	39.4	25.5	23.5	(31)	
		MCDB	15.4	18.7	34.3	42.5	53.1	74.6	75.6	69.3	57.4	40.9	26.3	24.6	(32)	
	10%	WB	10.0	13.6	29.1	34.9	43.1	62.7	65.4	61.0	51.0	35.7	21.9	18.5	(33)	
		MCDB	10.9	14.5	31.7	39.3	48.0	73.0	73.5	67.0	54.1	37.2	22.9	19.4	(34)	
	5% DB	MDBR	13.7	15.2	17.8	17.3	13.7	18.5	16.7	14.9	11.4	9.1	14.5	13.5	(35)	
		MCDBR	10.8	18.1	15.6	19.0	22.7	25.6	24.4	21.5	18.1	11.4	13.3	11.2	(36)	
		MCWBR	10.5	18.0	11.9	12.8	13.1	10.9	10.6	12.1	10.6	8.3	13.6	10.7	(37)	
		MCWBR	11.7	18.6	14.7	16.2	19.8	19.3	19.4	18.4	14.8	10.1	12.6	10.7	(38)	
(42) Clear-Sky Solar Irradiance	taub		0.234	0.284	0.326	0.349	0.367	0.383	0.429	0.408	0.347	0.299	0.216	0.226	(40)	
		taud	1.874	1.989	1.991	2.038	2.210	2.386	2.232	2.261	2.402	2.196	1.904	1.867	(41)	
	Ebn at Noon		89	186	230	257	266	265	245	238	233	184	94	28	(42)	
		Edh at Noon	10	22	34	41	38	33	38	33	23	18	9	4	(43)	
(44) All-Sky Solar Radiation	RadAvg		58	298	797	1399	1713	1786	1557	1048	601	239	91	22	(44)	
	RadStd		7	22	56	139	162	156	175	133	82	39	13	2	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (2 neighbors)		+1.76	N/A	N/A	N/A	N/A	N/A	-670	-819	N/A	N/A	

Nomenclature: See separate page